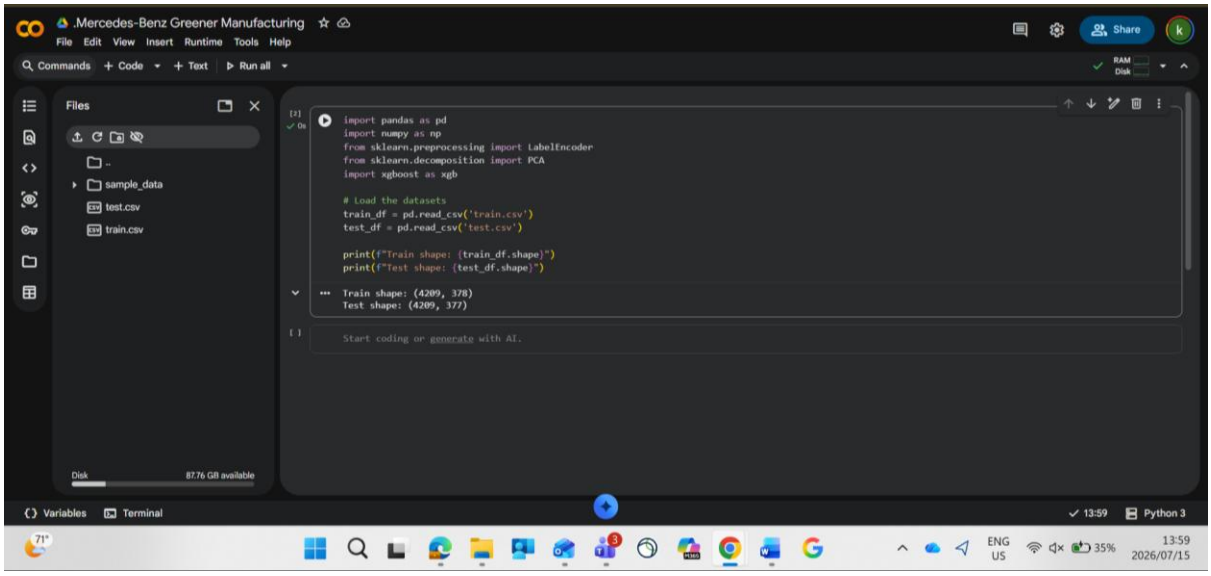
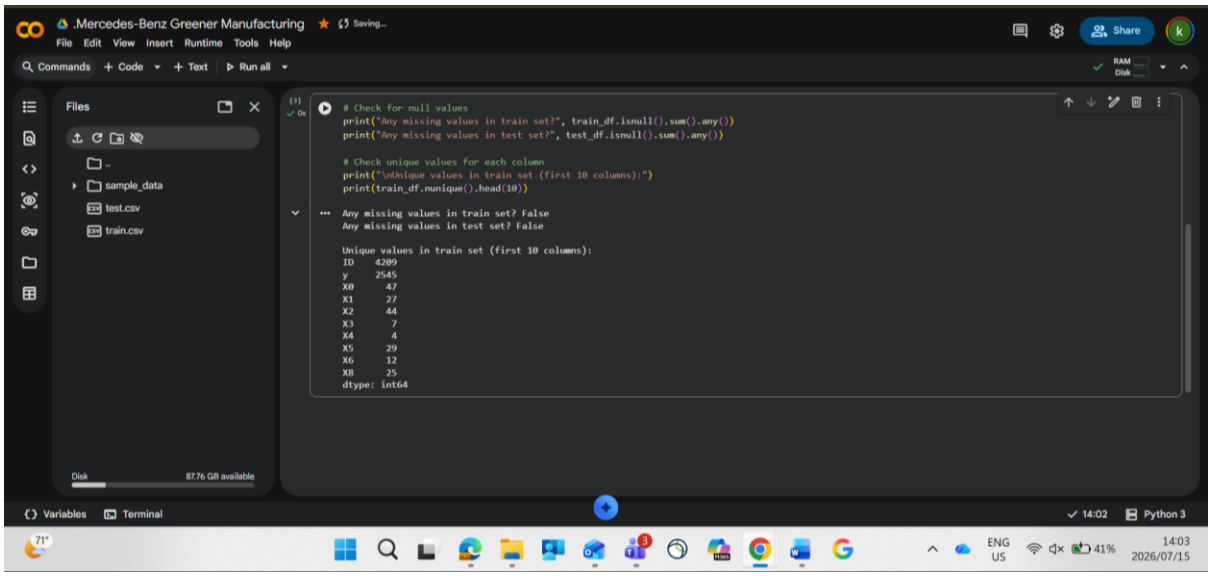
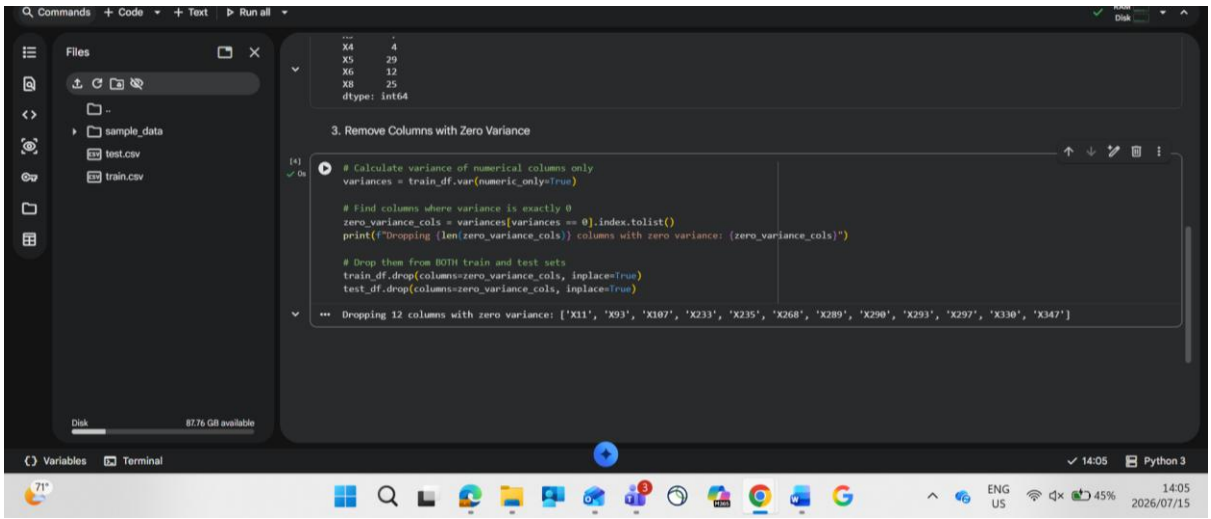
1. Initial Setup and Loading Data



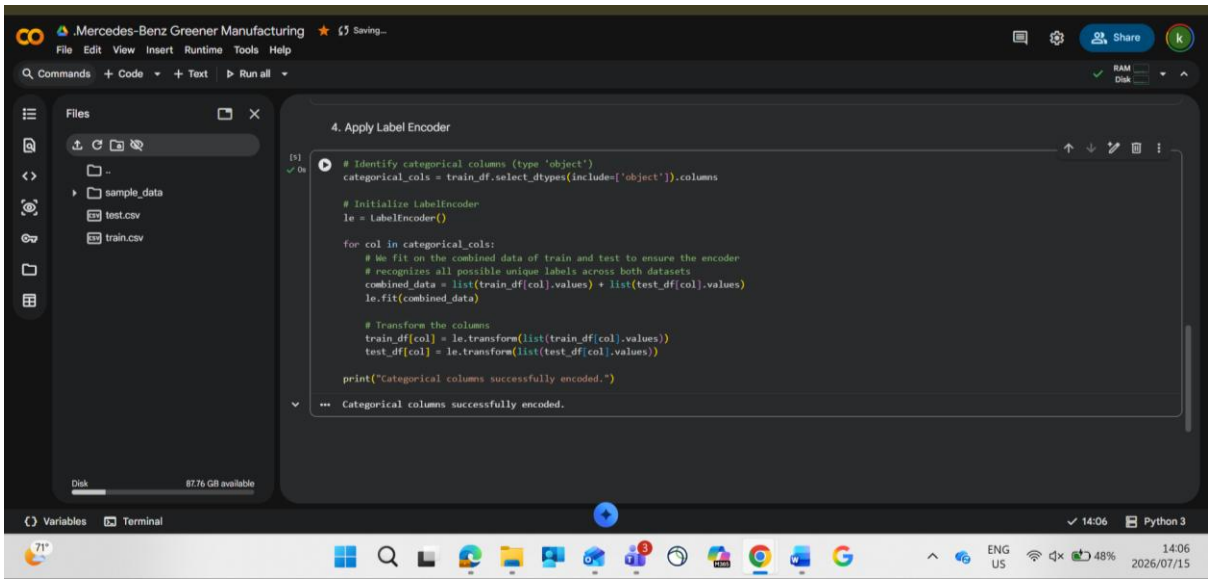
2. Check for Null and Unique Values



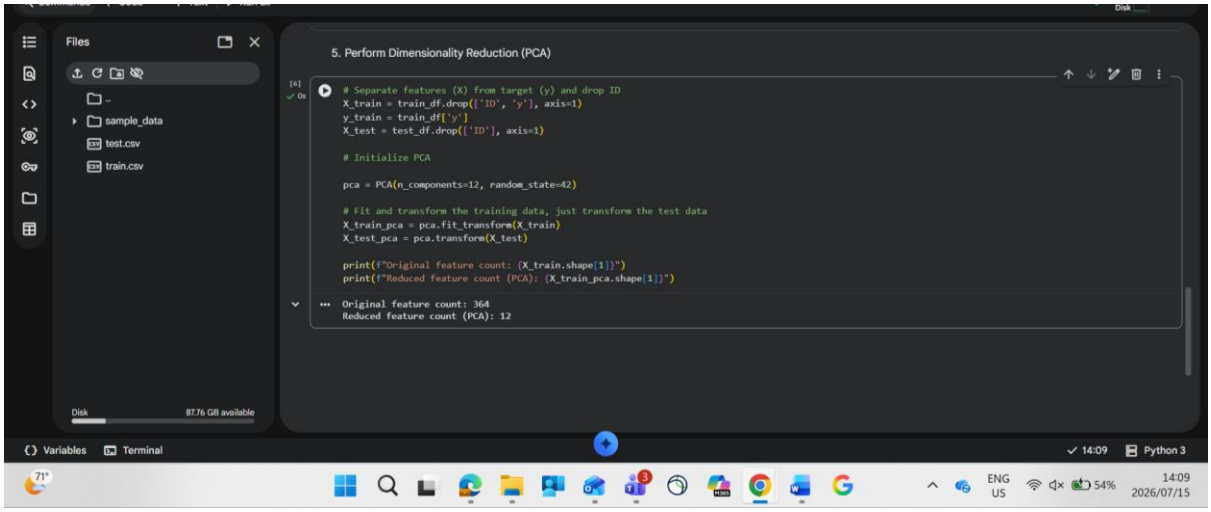
3. Remove Columns with Zero Variance



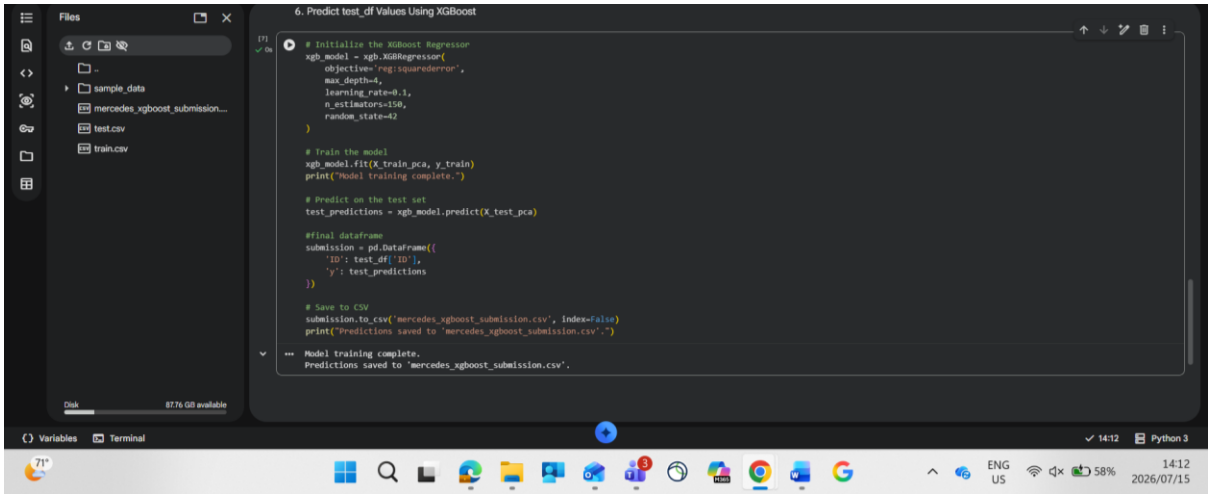
4. Apply Label Encoder



5. Perform Dimensionality Reduction (PCA)



6. Predict test_df Values Using XGBoost


Files
sample_data
mercedes_xgboost_submission....
test.csv
train.csv
Disk
87.76 GB available
6. Predict test_df Values Using XGBoost
Initialize the XGBoost Regressor
xgb_model = xgb.XGBRegressor(
objective='reg:squarederror',
max_depth=4,
learning_rate=0.1,
n_estimators=150,
random_state=42
)
Train the model
xgb_model.fit(X_train_pca, y_train)
print("Model training complete.")
Predict on the test set
test_predictions = xgb_model.predict(X_test_pca)
#final dataframe
submission = pd.DataFrame({
'ID': test_df['ID'],
'y': test_predictions
})
Save to CSV
submission.to_csv('mercedes_xgboost_submission.csv', index=False)
print("Predictions saved to 'mercedes_xgboost_submission.csv'.")
Model training complete.
Predictions saved to 'mercedes_xgboost_submission.csv'.
Variables
Terminal
14:12
Python 3
ENG
US
58%
14:12
2026/07/15